

Earliest Deadline First

```
#include <stdio.h>
#include <stdlib.h>

#define INFINITY 999999

typedef struct {
    int id;
    int arrival;
    int burst;
    int deadline;
    int period;
    int remaining;
    int current_deadline;
} Process;

int main() {
    int n;
    printf("1BM24CS361, Nitya R Swadi\n");

    printf("Enter the number of processes: ");
    if (scanf("%d", &n) != 1 || n <= 0) {
        return 1;
    }

    Process proc[n];

    for (int i = 0; i < n; i++) {
        proc[i].id = i + 1;
        printf("Enter arrival, burst, deadline, and period for Process %d: ", proc[i].id);
        scanf("%d %d %d %d", &proc[i].arrival, &proc[i].burst, &proc[i].deadline,
        &proc[i].period);
        proc[i].remaining = 0;
        proc[i].current_deadline = proc[i].deadline;
    }
```

```

printf("\nPID\tBurst\tDeadline\tPeriod\n");
for (int i = 0; i < n; i++) {
    printf("%d\t%d\t%d\t%d\n", proc[i].id, proc[i].burst, proc[i].deadline,
proc[i].period);
}

int simulation_time = 20;
printf("Scheduling occurs for %d ms\n\n", simulation_time);

int time = 0;

while (time < simulation_time) {

    for (int i = 0; i < n; i++) {
        if (time >= proc[i].arrival && (time - proc[i].arrival) % proc[i].period == 0) {
            proc[i].remaining = proc[i].burst;

            int current_instance = (time - proc[i].arrival) / proc[i].period;
            proc[i].current_deadline = proc[i].arrival + (current_instance * proc[i].period) +
proc[i].deadline;
        }
    }

    int min_deadline = INFINITY;
    int selected = -1;

    for (int i = 0; i < n; i++) {
        if (time >= proc[i].arrival && proc[i].remaining > 0) {
            if (proc[i].current_deadline < min_deadline) {
                min_deadline = proc[i].current_deadline;
                selected = i;
            }
        }
    }
}

```

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    if (selected == -1) {
        printf("%dms : CPU is idle.\n", time);
        time = time + 1;
    } else {
        printf("%dms : Task %d is running.\n", time, proc[selected].id);
        proc[selected].remaining = proc[selected].remaining - 1;
        time = time + 1;
    }
}

return 0;
}

```

OUTPUT:

```

C:\Users\Nitya\Downloads\rm X + v
1BM24CS361, Nitya R Swadi
Enter the number of processes: 3
Enter arrival, burst, deadline, and period for Process 1: 0 3 7 20
Enter arrival, burst, deadline, and period for Process 2: 0 2 4 5
Enter arrival, burst, deadline, and period for Process 3: 0 2 8 10

PID      Burst  Deadline  Period
1         3       7         20
2         2       4          5
3         2       8         10
Scheduling occurs for 20 ms

0ms : Task 2 is running.
1ms : Task 2 is running.
2ms : Task 1 is running.
3ms : Task 1 is running.
4ms : Task 1 is running.
5ms : Task 3 is running.
6ms : Task 3 is running.
7ms : Task 2 is running.
8ms : Task 2 is running.
9ms : CPU is idle.
10ms : Task 2 is running.
11ms : Task 2 is running.
12ms : Task 3 is running.
13ms : Task 3 is running.
14ms : CPU is idle.
15ms : Task 2 is running.
16ms : Task 2 is running.
17ms : CPU is idle.
18ms : CPU is idle.
19ms : CPU is idle.

Process returned 0 (0x0)   execution time : 23.368 s
Press any key to continue.

```